



Dissertation on

**Time Series Analysis to Forecast the Trend and
Seasonality of Region-wise Covid-19 active cases in an
Indian context**

Submitted in partial fulfilment of the requirements for the award of degree of

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in
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Submitted by:

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CERTIFICATE

This is to certify that the dissertation entitled

Time Series Analysis to Forecast the Trend and Seasonality of Region-wise Covid-19 active cases in an Indian context

is a bonafide work carried out by

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in partial fulfilment for the completion of 4th semester Capstone Project Phase - 2 (UE19CS651) in the Program of Study - Master of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Jan. 2021 – May. 2021. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 4th semester academic requirements in respect of project work.

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DECLARATION

We hereby declare that the Project Phase - 2 entitled “**Time Series Analysis to Forecast the Trend and Seasonality of Region-wise Covid-19 active cases in an Indian context**” has been carried out by us under the guidance of Prof.Nitin V Pujari, Dean-IQAC and submitted in partial fulfilment of the course requirements for the award of degree of **Master of Technology in Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester January – May 2021. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ABSTRACT

Time-series forecasting is the process of predicting future values based on the currently available values. Time-series forecasting has a lot of applications in many fields involving decision making and future forecasting.

Problem Definition

Time-series forecasting is also used in forecasting future conditions of an epidemic. Similarly, COVID-19 as well is an epidemic that can be effectively forecasted using time-series analysis. Hence time-series analysis has to be applied to COVID-19 data in India so that the results can be analyzed and used by individuals to plan precautionary measures and assess appropriate control measures to reduce the number of infections.

Goal

The forecast results obtained from the model are put up against the actual data of pandemic in the Indian context using certain evaluation metrics to determine the accuracy of the forecast. The accuracy of the model forecast can be improved by addressing the influence of various aspects that may affect the spread of the virus.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
1.	INTRODUCTION	01
	1.1 Time-Series analysis over the pandemic	04
2.	PROBLEM DEFINITION	06
3.	LITERATURE SURVEY	07
	3.1 Paper 1: COVID-19 Future Forecasting Using Supervised Machine Learning Models	07
	3.2 Paper 2: COVID-19 Predictions for India An Initial Attempt	10
	3.3 Paper 3: Predicting the growth and trend of COVID-19 pandemic using machine learning and cloud computing	12
	3.4 Paper 4: Trend Analysis and Forecasting of COVID-19 outbreak in India	15
	3.5 Paper 5: Spatial prediction of COVID-19 epidemic using ARIMA techniques in India	17
	3.6 Other References	18
	3.7 Existing Systems	20
	3.8 Conclusions	22
4.	PROJECT REQUIREMENTS SPECIFICATION	23
	4.1 Project Perspective	23
	4.2 Requirements	24

5.	PROPOSED METHODOLOGY	26
6.	SYSTEM DESIGN	28
	6.1 Design Considerations	28
	6.2 High Level System Design	30
	6.3 Low Level System Design	33
7.	IMPLEMENTATION AND PSEUDOCODE	36
	7.1 Dataset Preprocessing	36
	7.2 Data Science Models	38
	7.3 Test Strategy	42
8.	RESULTS AND DISCUSSION	44
9.	CONCLUSIONS AND FUTURE WORK	51
	9.1 Conclusions	51
	9.2 Future Work	52
	REFERENCES/BIBLIOGRAPHY	54
	APPENDIX A DEFINITIONS, ACRONYMS AND ABBREVIATIONS	
	APPENDIX B USER MANUAL (OPTIONAL)	

LIST OF FIGURES

Figure No.	Title	Page No.
6.1	High Level Design of the model	32
6.2	Low Level Design of the model	33
7.1	Rolling statistics plot to check stationarity of the data	39
7.2	ACF and PACF plots to determine p and q values	40
7.3	RNN model	42
8.1	Forecast over the first test interval by ARIMA model	45
8.2	Forecast over the first test interval by ES model	46
8.3	Forecast over the second test interval using ES and LSTM models	46
8.4	Forecast plots over the second test interval using ES and LSTM models	47
8.5	Future forecast of pandemic over Bengaluru Urban by ARIMA	47
8.6	Future forecast of pandemic over Bengaluru Urban by LSTM	48
8.7	Future forecast of pandemic over Bengaluru Urban by ES	48
8.8	Forecast plots of Chittoor district over first and second test interval by ES model	49
8.9	Future forecast of pandemic over Chittoor district by ES model	49
8.10	Forecast plots of Kolkata district over first and second test interval by ES model	50
8.11	Future forecast of pandemic over Kolkata district by ES model	50

LIST OF TABLES

Table No.	Title	Page No.
3.1	Paper 2: Analysis of the model predicted values with actual values	11
7.1	Processed COVID-19 Indian districts' time-series dataset	37
8.1	Quarterwise analysis of the dataset	44

CHAPTER 1

INTRODUCTION

COVID-19 popularly also called Coronavirus disease 2019 is an infectious disease caused by severe acute respiratory syndrome coronavirus2 (SARS-CoV-2). COVID-19 was declared as a pandemic by the World Health Organization on 11 March 2020.

The first case of COVID was detected in Hubei province, Wuhan city, China on November 17, 2019. COVID-19 causes severe fatal blockage in respiration and further leading to blood clots, hypercytokinemia, and sometimes death.

COVID-19 is contagious and spreads from one host to another through contact. It is primarily transmitted through the droplets of the sneeze, coughs, and exhales that can find its way to other people through the air or any other medium.

COVID-19 has taken the world by storm infecting a large number of people. Claiming over a million lives, COVID spread rapidly to all the other nations from China, and due to its mysterious and uncertain transmission, it was found safe if people stopped coming into close contact.

Hence upon WHO's assertion, various nations announced a complete lockdown in an attempt to cut down people's interactions. International travel and world-wide events were called-off or rescheduled for the future.

As a result, all the operations and businesses came to a halt causing the economy of the whole world to plummet. This caused a lot of loss for people and the government. It as well cost people their freedom making them vulnerable to mental problems.

It has even paved the way for the postponement or cancellation of events, widespread supply shortages worsened by panic buying, agricultural disruption, and food shortages. The effect of COVID-19 lockdown empirically witnessed a decrease in the emission of pollutants and greenhouse gases because vehicles and industries were non-functional.

In addition to complete lockdown, the WHO has insisted every individual follow the practices of social distancing, wearing masks, and sanitizing the surroundings frequently in an attempt to reduce the number of cases. While this helped a lot in limiting the flow of contaminants, there still has been a large number of cases nevertheless.

Countries highly affected by the pandemic were China, Germany, Spain, Italy, France, the USA, India, Brazil, and Russia which collectively recorded over half the number of universal cases. India recorded the second most number of COVID-19 cases just after the USA in accordance with the reports [1] obtained while framing the report. The infection varied from one country to another due to various factors such as prevention policies, climate, availability of medical facilities, population, and density of population.

The pandemic found its way to India in the mid of March 2020 and observed the condition of the other countries battling the pandemic as per the government advisory to shut down all the operations and enforce lockdown countrywide to avoid future widespread of the virus. Various factors such as the population density, lack of hygiene, and unavailability of appropriate health infrastructure in India posed a huge threat.

A lot of medical research has been carried out to synthesize a cure for the virus. The WHO along with the national bodies of all the nations and medical institutions all across the globe are investing huge amounts of money and time to prepare a vaccine.

Various vaccines are synthesized and many of them are fruitful increasing the immunity of the human body against the pathogens. Vaccines approved by Indian Council of Medical Research(ICMR) [3] are being administered to people at a very respectable pace.

The number of vaccine administrations increased and the number of cases reduced consistently for a period of three months during the first quarter of 2021 but the pandemic spread more vigorously than the first wave of the pandemic leading to more number of cases than that in the first wave. According to the current reports stated by [1], India has reported 1,53,26,624 cases and 1,80,577 deaths amidst the second wave of the pandemic as on April 20th 2020 at 4:30 PM.

Some of the other fields contributing research to stop the pandemic would be geographical research, the field of economics, technological research, statistical research, sociology, psychology, etc. There is a vast application of technology in combat against the virus.

Some of the applications offered by the technology fields are data analytics to forecast the pandemic, machine learning methods to identify disease in a patient (using deep learning imagery in order to determine the pathogens from x-ray images of patients' chest [6]), integrated machine design for replacing humans in cases of virus diagnosis and delivery applications, virus tracking applications and biomedical applications. The attempt of Artificial Intelligence to combat coronavirus is explained in [5].

Although the synthesis of medicine was fruitful it would at least take half a year for the vaccine to be available to cover the entire population as per the respective government directives. In order to address the issue of efficient distribution of the vaccine with the most minimal loss of lives, it would be helpful to know where the necessity of the vaccine would be maximum beforehand.

To predict the COVID-19 virus it is required to track the interaction of each and every person and keeping track of people's movement involves the intrusion of privacy and difficulty in contact tracing. Thus an alternative would be to develop inferences from the available statistical data over the pandemic.

1.1 Time Series analysis over the pandemic

Time series analysis is a technique that deals with the study of the trend of any statistically measured variable with respect to time. Tracking an activity to develop an understanding of it can be best incorporated using time series. Time series are most widely used in the applications of weather, stock price, retail, and economics.

Plotting time series, one can visually look at the trend and determine future distribution as well. Time series forecasting makes use of the existing data plotted against time in order to determine the data values for the next time frame.

Time series analysis consists of various methods to extract meaningful statistical inferences from the data. Various data science models are available to make forecasts from the available time-series data. Some of the well-known models are Linear Regression (LR), Moving Average (MA), Long-short term memory (LSTM), etc.

Statistical models have been used for the purpose of forecasting and decision making for a long time. Applying effective data science models over sufficiently legit data often gives remarkable results. In history, time-series analysis has been used in the field of disease forecasting and has been proven to be quite effective as observed in [7], [8].

Article [9] explains the ways and techniques to utilize time-series data and data science to monitor and control disease spread. It is always necessary for the data to be legitimate to obtain a good forecast of the future.

The stated problem at hand requires a similar kind of approach to model the growth of the virus. This study is as well aimed at utilizing data science models in order to make a time-series forecast upon the trend of the growth of the virus in an Indian context.

CHAPTER 2

PROBLEM DEFINITION

“Time Series Analysis to Forecast the Trend and Seasonality of Region-wise Covid-19 active cases in an Indian context”

Forecasting the trend of the spread of the COVID-19 pandemic, so that it can assist people, medical facilities, and control agencies in making informed decisions. The one method proposed in the review would be the usage of statistical data of the COVID-19 virus-infected patients across India and the usage of data science models to develop a time-series analysis of the derived time-series data.

COVID-19 is a relatively new domain of research and not much research is dedicated to it. The attempts towards forecasting the growth of the pandemic are enlisted and elaborated in the Literature Survey(section 3). Various models of forecasting are used to forecast the number of cases accurately and also determine which model gives the best forecast results.

The goal is to build a data science time-series model that efficiently forecasts the future pandemic conditions with an estimated higher accuracy. Accuracy can be improved by studying and analyzing the association of the parameters that influence the spread of the pandemic.

CHAPTER 3

REVIEW OF LITERATURE

The use of data science models to forecast the future trend of the COVID-19 virus will bear quite an appropriate result for efficient decision-making. The accuracy of the prediction is highly dependent on the quality of the dataset available. While this is not the first time using statistical time series forecasts to assist pandemic decision making, the following section illustrates some of the applications.

3.1 COVID-19 Future Forecasting Using Supervised Machine Learning Models

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KFUEIT, Taibah University, Islamia University, Kunsan National University, Yeungnam University

Indexing: IEEE 25-05-2020

The study [10] demonstrates the power of Machine Learning models to foretell the count of upcoming cases of the COVID-19 virus. The authors state the usage of the existing and renowned ML models to make an efficient forecast of the future number of cases. The study makes forecast utilizing the following forecast models,

-
- Linear Regression (LR)
 - Least Absolute Shrinkage and Selection Operator (LASSO)
 - Support Vector Machines (SVM)
 - Exponential Smoothing (ES)

The author further takes the liberty of evaluating each of the models' efficiency and comparing against one another and determining which gets the most accurate result. The results are compared against one another by using certain performance metrics, R-squared score (R^2 score), Adjusted R-squared Score (R^2_{adjusted}), mean square error (MSE), mean absolute error (MAE), and root mean square error (RMSE).

Implementation

The models are applied over the data obtained from the GitHub repository by the Center for Systems Science and Engineering, Johns Hopkins University [15]. The future forecast is made for the time frame of the next 10 days by utilizing the number of new positive cases, the number of deaths, and the number of recoveries obtained from the dataset.

The data is preprocessed to estimate the global statistics of the number of confirmed cases, deaths, and recoveries in the past since the pandemic started. Then the cleaned dataset is divided into training data of 56 days and test data of 10 days.

The learning models mentioned earlier are applied over this data and have been trained on the days and newly confirmed cases, recovery, and death patterns. These models have been evaluated and compared based on the evaluation metrics stated above. The tests were

conducted for 4 different time-frames in order to determine the level of performance of a model based on the amount of data fed.

Results and Conclusions

The experiment resulted in smooth results in forecasting the future number of patients that would be infected by COVID-19. The model detected a greater threat to society by the virus. Considering a smaller time frame of training data of only 26 days it was found that all the models performed badly due to the availability of limited data except for Exponential Smoothing.

Iteratively considering a larger dataset each time up to all the 66 days of data, the other models' performance was found to drastically increase in the order of LR, LASSO, and SVM. Applying the models over all the data put together, all the models performed fairly while SVM still struggled behind.

The performance of the models was ranked as ES, LR, LASSO and finally, SVM and the author thus concluded that ES provided the best forecast results both in short-term and long-term predictions. This model concludes by stating the importance of the amount of data available to conduct research.

3.2 COVID-19 Predictions for India An Initial Attempt

Abhijit Das

IIT Kharagpur

Indexing: Research report papers - IIT-Kharagpur 13-06-2020

This paper [11] documents the findings of research conducted at IIT in an attempt to make predictions over the COVID pandemic in India. The work presents a way to make predictions over the pandemic risk based on the cumulative count of infections.

The model developed here presents its pandemic forecasts in terms of the rate of spread, peaking date, and ending date and is applied to make forecasts both on the state-level data and nation-level data.

The author urges that in order to make accurate forecasts which require close attention to the movement of humans as they can not be treated as random gas molecules. But this results in intrusion of their privacy and implementing such a system for a population as large as India would be very expensive. Hence the next best alternative is the usage of the pandemic spread data that represents useful information of the propagation of the virus.

Implementation

The daily available counts are summed in order to obtain a cumulative total number of cases C_N over a period of n days. Count for consecutive days C_i is used to determine the best estimates of the rate of infection r and the total number of cases K .

Initially, K and r values are randomly selected and the best estimates are found by running a non-linear regression. These values can, in turn, be used to verify what N value gives the best estimates i.e determining the time-frame to consider in order to get the best results.

The author utilizes logistical models to fit the available daily counts of infection data. RMS error is utilized in order to obtain the best estimates for the model. Moving average technique is used to smoothen daily noises. The results for the model are obtained for the whole nation and states of Maharashtra and West Bengal which are compared with the actual infection data which was found later on.

	Date	N	K	r	E_{rms}	End on	Actual Cases A	A - K	MAE
IN	03-06-2020	46	467353	0.069	1408	13-09-2020	216876	250477	286261.7
	04-06-2020	47	482624	0.069	1456	14-09-2020	226723	255901	
	05-06-2020	48	496703	0.069	1498	15-09-2020	236195	260508	
	06-06-2020	36	534399	0.067	1546	20-09-2020	246603	287796	
	07-06-2020	49	555637	0.067	1597	21-09-2020	257485	298152	
	08-06-2020	50	564909	0.066	1612	22-09-2020	266021	298888	
	09-06-2020	51	572899	0.066	1625	23-09-2020	276002	296897	
	10-06-2020	51	587702	0.066	1646	24-09-2020	287158	300544	
	11-06-2020	52	597997	0.066	1674	25-09-2020	298293	299704	
	12-06-2020	52	623349	0.065	1703	27-09-2020	309599	313750	
MH	03-06-2020	44	123813	0.083	469	20-08-2020	72300	51513	46343.9
	04-06-2020	44	128012	0.082	468	22-08-2020	74860	53152	
	05-06-2020	45	128113	0.082	465	22-08-2020	77793	50320	
	06-06-2020	45	128113	0.082	465	22-08-2020	80229	47884	
	07-06-2020	47	129579	0.081	471	23-08-2020	82968	46611	
	08-06-2020	48	130493	0.081	477	23-08-2020	85975	44518	
	09-06-2020	49	131091	0.081	479	23-08-2020	88528	42563	
	10-06-2020	49	132546	0.081	498	24-08-2020	90787	41759	
	11-06-2020	42	136580	0.080	550	26-08-2020	94041	42539	
	12-06-2020	49	140228	0.078	622	28-08-2020	97648	42580	
WB	03-06-2020	39	12882	0.077	135	02-09-2020	6168	6714	15133.2
	04-06-2020	40	14682	0.075	141	07-09-2020	6508	8174	
	05-06-2020	41	16826	0.073	146	12-09-2020	6867	9959	
	06-06-2020	42	19247	0.072	151	16-09-2020	7303	11944	
	07-06-2020	43	21872	0.071	155	20-09-2020	7738	14134	
	08-06-2020	37	27874	0.069	156	29-09-2020	8187	19687	
	09-06-2020	38	29406	0.068	156	30-09-2020	8613	20393	
	10-06-2020	39	29382	0.068	155	30-09-2020	8985	20397	
	11-06-2020	40	29365	0.068	154	30-09-2020	9328	20137	
	12-06-2020	41	29561	0.068	153	01-10-2020	9768	19793	

Table 3.1: Analysis of the model predicted value with actual data

Results and Conclusions

The results obtained in this study are analyzed and empirically evaluated to check for its legitimacy. The results obtained by the model as represented in the table shown in table 3.1, makes a very poor forecast while it is compared to the actual data.

It can be seen on any particular day, the predicted cases represented by K is more than twice as much as the cases actually seen. The author further predicts the end of the pandemic and it can be witnessed that the virus still prevails in society. Hence the techniques used are unreliable and flawed models or data might result in disastrous strategy planning.

3.3 Predicting the growth and trend of COVID-19 pandemic using machine learning and cloud computing

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Obtaining the best forecast employs the usage of the available resources and technology in a very efficient manner. The authors propose the utilization of cloud computing technology with machine learning methods to improve the model's availability and timeliness.

The paper [12] describes the deployment of the standard machine learning forecast model over a cloud framework in order to make the model available anytime anywhere. Many researchers

believe the coronavirus to follow a general normal distribution and model by [42] mentioned by the authors as well do so.

The authors describe that the previously archived reports over SARS-CoV-1 virus [16] which was witnessed in the year 2003, was found to follow a General Inverse Weibull distribution better than Gaussian distribution.

The authors claim that the growth distributions of the pandemic follow an Inverse Weibull distribution [17] rather than the normal distribution and present an iterative weighting for fitting generalized Inverse Weibull distribution. By this study, they claim to have obtained a better fit using this model.

Implementation

The study proposes a framework to deploy the machine learning models on cloud data-centers to provide a fail-safe computation and quick data analysis. This enables continuous uninterrupted access to the model over the internet from any part of the world.

For the case study, they have used 3 instances of single-core Azure B1s virtual machine with 1GiB RAM, SSD storage, and a 64-bit Microsoft Windows Server 2016. They have used the HealthFog [18] framework leveraging the FogBus [19] for deploying multiple analysis tasks in an ensemble learning fashion to predict various metrics, like the expected number of medicinal facilities to assist patients.

The data is obtained from [36] and additionally from [37]. Levenberg-Marquardt (LM) or damped least square method for curve fitting (DLS) is used to predict parameters of the model. The study proves that there lie many outliers and noise in the initial COVID data which makes it hard to accurately predict.

Thus the authors propose an iterative weighting strategy which is called Robust fitting. The results obtained are gauged using MSE, MAPE, and coefficient of determination. Population density, average and median age, weather conditions, health facilities, etc. are also to be integrated for enhancing the accuracy of the predictions.

Various models were considered and the best performing 5 distributions were selected which are Weibull distribution, Gaussian distribution, Beta 4 distribution, Fisher-Tippett distribution, and Lognormal distribution. The performance of each of them is put up against each other and they conclude that the Weibull distribution fits the best.

Results and Conclusions

The results are obtained over the whole world over all the major countries and a global heatmap is generated representing the country-wise distribution of risks. By visual and empirical analysis over the actual coronavirus graph data obtained from [44], the trend of the virus spread does seem to follow a Gaussian distribution and not inverse Weibull distribution. However, the conclusions drawn by the authors with regards to their observations are found to not match the actual distribution of the time series data.

3.4 Trend Analysis and Forecasting of COVID-19 outbreak in India

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The paper [13] attempts at predicting the extent of the spread of the virus and the effects of it before the virus caused a large spread disease in India. This is a relatively early paper when compared to the other research papers available as it attempts at making predictions of Indian conditions before the virus was found in large numbers.

It attempts to describe the readiness level of the nation and the effect of the epidemic over the nation based on the then-available data and the worldwide data. The study also discusses the regional analysis of Indian states and presents the preparedness level of India against this virus.

The author mentions the usage of other nation's pandemic data to make conclusions about our country and this study is limited to analyze and make forecasts only for India. The author further states the high risks as uncontrolled pandemic might cause a lot of fatalities in India due to the high population.

Implementation

The author uses the ARIMA model and Exponential smoothing methods to forecast the pandemic growth in India in the early stages. The model is applied over the dataset obtained

from the repositories of John Hopkins University [39], comprising 60 days of training data and 30 days of test data. The model makes a forecast over the total and the new number of infected cases and death cases for the next 30 days.

The analysis is divided into 3 phases,

- Trend analysis of the number of infected cases and death cases assessed for India.
- State-wise analysis conducted for India showcasing various penetration levels and mortality rates for different states and union territories in the country.
- Comparative analysis of India with other SAARC nations to assess the level of COVID-19 spread in the south Asian region.

Results and Conclusions

The results for all the above 3 mentioned phases are schematically illustrated in the research paper along with the insights gathered from them. Forecasting the virus spread over the nation it is witnessed that the increase in the number of COVID cases in India is steep and has to be addressed early before the pandemic infects much of the population. The lack of assurance of the forecast in this research can be seen due to the lack of data and hence it can be concluded that lack of data often leads to improper judgments.

3.5 Spatial prediction of COVID-19 epidemic using ARIMA techniques in India

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The accuracy of any forecast can be improved by using the factors influencing the trend and this study also promises better accuracy by using the location information. In the assessment of risks of any disease especially in the pandemic of COVID-19, geographical knowledge plays a vital role.

The spatial distribution of pandemic risk is carried out by utilizing a weighted overlay in the GIS platform. The study proposes the usage of real-time maps to make it look more comprehensible and this can help sooner assessment of threats.

The spatial aspect refers to the population density, number of cases, and the disease status over the region. The regions vaguely represent the districts and the whole nation's map is divided into 4 categories based on the risk which are low, medium, high, and very high. ARIMA model is deployed to make forecasts which are measured by using MAE and RMSE.

Implementation

This study [14] demonstrates certain effective short-term prediction models to estimate the number of possible events. The data used by the model was dated from 30 January 2020 to 26

April 2020 for training purposes and further data dated from 27 April 2020 to 11 May 2020 was made use in forecasting samples.

The dataset was obtained from [38] of which they utilized a variable holding the cumulative confirmed cases in order to make predictions. The stationarity of the time series was checked by performing the Augmented Dickey-Fuller test and the data was made stationary by applying log-transformations and obtaining the differences.

Auto-correlation function and partial autocorrelation functions were applied to determine the model parameters. The study resulted in plotting the high-risk regions and stated that southern and western India are more vulnerable. The study also states a higher risk in the older neighborhood than the newer ones.

Results and Conclusions

The ARIMA model gives the best fit for COVID forecast when compared to the other models and makes forecast possible for non-stationary time series data as well. Consideration of the spatial aspect of the pandemic spread makes the forecast more precise and accurate.

3.6 Other references

There are various other aspects of obtaining forecasts of the pandemic apart from the ones mentioned above. Some of the other attempts at forecasting the virus are mentioned in this

section. The proposed application seeks inspiration from the models elaborated in this section and the previous one.

Deep learning and data science are majorly used in forecast applications. One such solution of using Artificial neural networks for predictive surveillance of COVID-19 is presented in [20]. [21] introduces the way of using Data science models to forecast and combat the COVID virus.

[22] proposes a hybrid model that promises high accuracy by integrating Long-Short Term Memory (LSTM) and Natural Language Processing (NLP). The model refines the forecast results by extracting insights from the online available news articles using NLP.

[23-25] present different prediction models in India. These models are derived based on the standard epidemiological models. [26] uses Ricatti differential equations to develop a data-guided detection in the USA and [27] utilizes the SIQR epidemiological model to make a time-window analysis of COVID-19 outbreak in Italy. Similarly, a case study over the pandemic crisis in Canada is presented in [28].

Considering the location attribute for the pandemic spread improves the accuracy of the model. This can be realized in ways depicted in [29]. The study elaborates on the ways to develop geographical detectors to obtain a decision tree model. [35] presents a way of closely monitoring the pandemic spread at the personal level.

Social media generates a lot of data during the time of pandemic which can assist in forecasting. Most information found on social media is not relevant and extracting the needed data from data trash is tedious. Like [22] uses news articles, [30] uses Naive Bayes and support vector

machines to draw situational data from social media. [32], [33] present a case study over the usage of social media for the ebola pandemic.

The more precise picturization of the movement of the virus can be tracked by closely looking at the human activity as suggested in [34] which uses the existing cellular networks to identify the regions at risk due to coronavirus. This study conveys the usage of handover and reselection technology to achieve so, eliminating the need for new applications.

A similar study is presented in [31] which speaks of using a smartphone application that collectively shares the data of potential interest to the rest of the society. This study was only confined to the COVID-19 data of Spain.

3.7 Existing Systems

With the rapidly spreading virus infecting a large number of people over the world there are a large number of data science researches being carried out with the aim of forecasting the spread. Some of the noteworthy examples are

COV-IND-19 Shiny App

Dr. Bhramar Mukharji, a professor at the University of Michigan with a doctorate in statistics has led a group of researchers to use standard epidemiologic models to do a situational assessment of the coronavirus crisis in India as described in [40].

The research did not follow the standard academic view of indulging paper works and rather implemented an agile development of adhering to data and making forecasts. However the results are claimed to be not too much perfect, it can be accessed at [41].

COVID-19 Projections

This web-based application makes use of Machine learning models to foresee the spread of the epidemic and visualize them graphically. This model is based on the standard epidemiological SEIR [43] model and makes predictions for most of the nations in the world. The model utilizes the number of daily deaths extracted from data obtained from John Hopkins University [39]. The model can be accessed anywhere anytime over the web [42] and is updated regularly.

Data-Driven Innovation Labs

This is a research facility [44] as a part of the Singapore University of Technology and Design (SUTD). They have attempted to build a forecast model using machine learning over the standard epidemiological SIR model. The research claims the coronavirus infection data to follow a Gaussian distribution.

There are other various projects carried out under this domain by various people from different parts of the world and thus in India as well. There are projects undertaken in various Indian universities and some carried out at the Indian Institute of Technology as well [38].

3.8 Conclusions

All the papers have a common agenda of forecasting the extent of the growth of the virus in order to assist effective enforcement of facilities and control policies. They only differ in the approach towards it in terms of data source, the model used and the parameters considered.

- Statistical analysis of data presents the trend of the virus based on behavioral history and is not completely certain about the predictions.
- Despite certain inaccuracies found, statistical analysis of data can bear quite appropriate results to assist the decision-making process.
- Accuracy of data is important for the model to derive the right results and the data has to be huge and available from a trustworthy source. With legitimate data at hand statistical analysis can provide an idea of the situation hence aiding decision making.
- The inclusion of additional features can improve the fit of the model. Some of these influencers are the density of population, age of a patient, availability of proper infrastructure, medical background of a patient, etc.

The models discussed in this section give an idea of what affects the forecast result and prioritization of different aspects in order to obtain the most accurate results possible. Some of the models give decent results while some go terribly wrong. Hence the prime goal of the model being developed in this project is to make forecasts by implementing all the positive points of the existing models to obtain forecasts with higher accuracy.

CHAPTER 4

PROJECT REQUIREMENTS SPECIFICATION

This section conveys the requirements for the implementation and development of the COVID-19 forecast system.

4.1 Project Perspective

The application proposed attempts to make forecasts over the future of the COVID-19 pandemic based on the statistical data. The project utilizes concepts of Computer Science, Data Science, and Statistics to achieve the goal. Hence machine learning-based software requirements and statistical data requirements can be postulated.

Forecasting models using statistical data are proven to be crucial for aiding decision-making for better enforcement of medical and health facilities, social awareness, and government control policies. This work aims at forecasting Covid-19 to assist decision-making.

Project Features

The application uses data-science models over statistical epidemic spread data in order to make a time-series analysis over the trend of growth of the pandemic and forecast the future

behavior of the virus. The application is aimed to provide more accurate forecasts with the help of available raw datasets hosted at [1].

4.2 Requirements

Functional Requirements

- Charts visualizing the trend of total infection cases, active cases, and total recoveries till date and for a user-specific timeframe.
- Representation of the number of total infected cases forecasted by the model.
- Region-based COVID-19 data graphs.

Hardware Requirements

Intel 6th gen core i5 / AMD Ryzen 5 and above

8GB DDR-4 RAM and above

Nvidia GTX 960 / AMD Radeon Rx 460 and above

Appropriate cloud services on a need basis

Software Requirements

Python 3.x and above

Latest Anaconda framework

Operating Environment

Processor: Intel Core i3 7100

RAM: 4 GB DDR4

GPU: AMD RX 470 4GB GDDR5

Platform: Anaconda Navigator 1.9

Operating System: Ubuntu 20.04 LTS

Non-Functional Requirements

- The application has to promise the reliability of the forecast for which it is as important for the data to be as reliable as the model. The data has to be obtained from a reliable and reputable source.
- Timeliness is the most important aspect as outdated data can cause inaccurate predictions which can lead to improper judgments.
- These attributes govern the effective and efficient forecast power of the application.

CHAPTER 5

PROPOSED METHODOLOGY

The goal of the project is to analyze the dynamics of the spread of the COVID pandemic and build a forecast model which can foretell the number of cases that could occur in the future. As explained in the review of literature section, statistical models have been used for forecasting pandemics for a long time and there are certain existing models to forecast COVID cases as well. The study aims at utilizing such statistical models to obtain pandemic forecasts and compare the accuracy of each such model.

The proposed methodology makes use of three different statistical models to forecast the number of COVID cases. These models are

- Exponential Smoothing(ES)
- Autoregressive Integrated Moving average(ARIMA)
- Long Short-term Memory(LSTM)

Forecast results and implementational details of each of the models is explained in the Implementation(), System Design(), and Results and Discussion() sections.

The raw data obtained from [1] is cleaned and processed by segregating the data into respective districts and arranging them chronologically. These chronologically arranged records contain year-long data consisting of dates and the number of active cases on the respective day. The dataset is then made model ready and the models are fit using this dataset.

Certain evaluation test metrics are established and the results obtained from the models are put up against each other and compared with the actual number of cases to determine which model yields the best results and how accurate the forecast actually is.

CHAPTER 6

SYSTEM DESIGN

This section specifies the high-level and low-level designs of the project and insights into the components of the application and how each of them functions. The document also briefs out a step-by-step procedure from data processing to visualization of results.

6.1 Design Considerations

- The forecast is done over the COVID-19 pandemic spread in India using the available statistical data while these forecasts obtained will be depicted in terms of charts.
- The forecast has to be accurate as it would be crucial while using it to neutralize the effects of the virus.
- The framework has to implement an efficient structure to handle the tabular data and has to be unambiguous.

Architecture Choices

There are a number of statistical models that can be used to forecast a statistical outcome like linear regression, lasso regression, ARIMA, etc. All models give similar results for input data and hence it is important to transform the data into a form that can be input into the model. The document deals with explaining the preprocessing and structuring of the COVID-19 data and as well discusses the probable statistical model to be used.

Constraints, Assumptions, and Dependencies

- The application at hand requires data that is subject to proprietary ownership. Hence the dataset used in the application development is open source.
- The forecast made by the model is highly dependent on the awareness of people and other features like social distancing, enforcement of prevention policies, climate, and availability of proper medical resources.
- Probabilistic models make assumptions based on the current trend of the pandemic and are prone to errors due to a sudden spike in the number of cases.
- Hence the model is applicable in an environment with social distancing enforced and believed that efficient hospital and patient care services are available.

Risks

The application makes forecasts based on the statistical data available over the pandemic and this does not address the actual events that happen in the physical world and the measures taken to prevent the spread. Since it is highly dependent on the data, there are chances of forecasts being inaccurate which can lead to false assumptions in making decisions.

6.2 High-Level System Design

The application is proposed to have time-series graphs plotted to represent the pandemic spread over the past and representing what the current scenario is. One could make a guess upon what the state in the near future could be looking at the plotted trend. Region-wise trends will as well be plotted and represented.

The plot of the future spread of the virus shall be determined by the model and represented graphically. For forecasting the number of future cases, the model uses the count of daily active cases, and the forecast is made only for India. The most accurate forecast can be promised only for a time frame of 15 days and hence the forecast can be obtained for the following 15 days from the last entry of the dataset.

Further, the model is aimed at determining the best volume of the dataset to be used and the best time frame to forecast the pandemic spread with accurate and respectable results. The

model described here is a data science application and hence there is no user interface for the analysis made but only a minimalistic graphical representation of the pandemic trend.

Platform Overview

The project is implemented in Python language using the anaconda framework. The dataset is handled by using the pandas library[4] of python which provides a very efficient way of handling tabular data. The models for forecasting will be implemented using respective standard python libraries. The structured data is then fed into these implemented models.

Dataset Design

The raw dataset is obtained from [1]. The repository contains a vast number of datasets of which the time-series dataset of the pandemic distributed over all the districts of the nation is extracted. A representation of the procedure of dataset processing is displayed in figure 6.1.

The district-wise time-series dataset represents the day-to-day update of the pandemic cases in all districts of the nation. The features of interest in this dataset are Date, State, District, the number of Confirmed cases, the number of recoveries, and the number of deaths. The data is segregated into respective districts and are combined to form a chronological time-series of the pandemic data of that district over the period of a year.

The data extracted constitute a total of 710 districts' data of the number of covid cases over a term of one year resulting in about 2,60,000 records of data. This data is handled and stored in python dataframes.

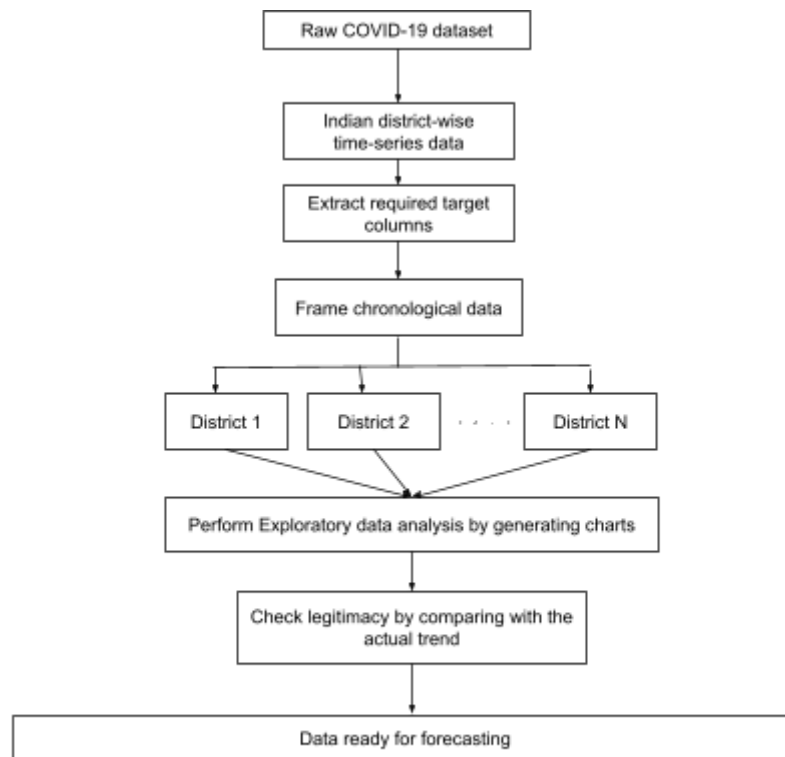


Figure 6.1: High-Level Design of the model

Report Layouts

The goal of the project is to generate charts and reports depicting the pandemic forecast. The reports generated by the model that represents the pandemic forecasts shall represent the count of the number of daily cases in the future along with the other data. The forecast data is distributed into districts and each table is associated with a district, state, and country.

Probable Models

The dataset which is transformed can now be fed into a machine learning model to obtain forecasts. The promised forecast results are inclusive of the effect of the features that influence the spread of the virus. The data is fed into 5 different models and each of their performances is put up against each other to determine the best fitting model. The probable models to be used to make forecasts are Linear Regression, Exponential smoothing, and the ARIMA model.

6.3 Low-Level System Design

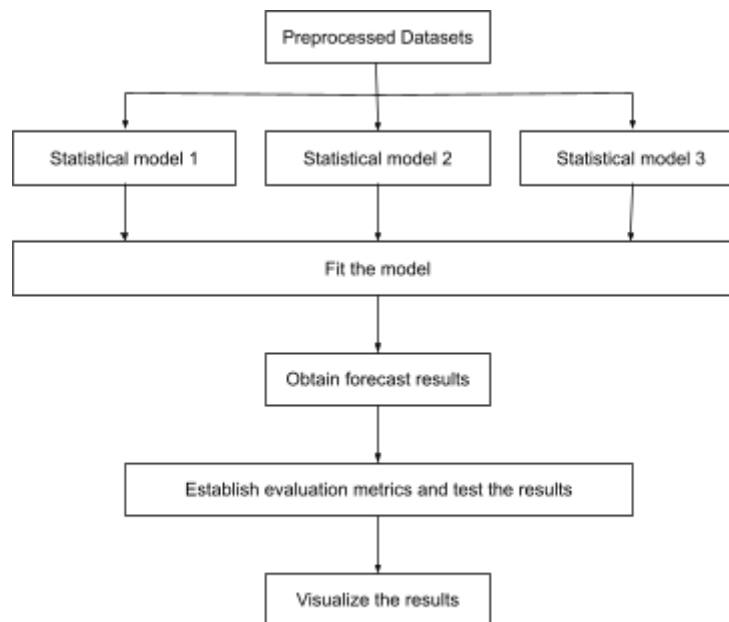


Fig 6.2: Low-level Design of the model

The proposed methodology makes use of three different statistical models to forecast the number of COVID cases in the future. The dataset is processed accordingly and the models are fit using this dataset. Certain evaluation test metrics are established and the results obtained from the models are put up against each other and compared with the actual number of cases to determine which model yields the best results and how accurate the forecast actually is.

Algorithm and Pseudocode

The three statistical models to be used for the forecast are Exponential Smoothing(ES), Autoregressive Integrated Moving average(ARIMA), and Long Short-term Memory(LSTM).

Exponential Smoothing

Exponential Smoothing(ES) is a simple yet powerful technique that makes use of the exponential window function for smoothing time series data. It is based on the assumption that the relevance of training data decays and becomes less important over time. Hence the older data will have a smaller influence over the model than the newer data. It is a univariate time series point.

$$s_0 = x_0$$
$$s_t = \alpha x_t + (1 - \alpha)s_{t-1}, \quad t > 0$$

where α is the *smoothing factor*, and $0 < \alpha < 1$.

ARIMA

ARIMA model, short for Auto-Regressive Integrated Moving Average, is a class of models that analyzes a time series making use of its past values. ARIMA model is widely used for forecasting

applications due to its capability of handling non-stationary data. The ARIMA model is obtained by integrating Auto-Regressive and Moving Average models and is represented by the equation

$$\text{ARIMA}(p,d,q)$$

where

- p is the number of autoregressive terms,
- d is the number of nonseasonal differences needed for stationarity, and
- q is the number of lagged forecast errors in the prediction equation.
- The AR model is the one where the output depends on its own lag while the MA model alone is based on the lagged forecast errors.

Long Short-term Memory

Long Short Term Memory(LSTM) is an artificial recurrent neural network(RNN) architecture that can be utilized in the applications of classifications and predictions. LSTM consists of feedback networks that serve the backpropagation purpose. The algorithm recursively uses the previous 'n' instance values to forecast the next value.

CHAPTER 7

IMPLEMENTATION AND PSEUDOCODE

7.1 Dataset Preprocessing

The dataset used for the project is district-wise time-series data of the spread of the COVID-19 virus. This dataset is obtained from [1]. The dataset is processed according to the directions specified in the dataset design section(section 6.2).

The district-wise time-series dataset consists of infection data of each district of the nation arranged chronologically. The dataset is loaded and the target attributes are extracted. The target columns represent the number of confirmed cases, recovered cases, and deceased count. The goal is to make a day-to-day representation of this distribution for each of the districts as represented in table 7.1.

Dataframes are derived to hold infection data of a specific state from the base dataframe and district-wise segregation is performed over this. A dictionary of dataframes is framed that holds the name of the district as the key and the respected day-to-day data as the value that belongs to the same state. This processed dataset shall be used to make forecasts using data science models.

Date	State	District	Confirmed	Recovered	Deceased
2020-04-20	Karnataka	Udupi	DNA	DNA	DNA
2020-04-21	Karnataka	Udupi	DNA	DNA	DNA
2020-04-22	Karnataka	Udupi	DNA	DNA	DNA
2020-04-23	Karnataka	Udupi	DNA	DNA	DNA
2020-04-24	Karnataka	Udupi	DNA	DNA	DNA
2020-04-25	Karnataka	Udupi	DNA	DNA	DNA
2020-04-26	Karnataka	Udupi	3	3	0
2020-04-27	Karnataka	Udupi	3	3	0
2020-04-28	Karnataka	Udupi	3	3	0
2020-04-29	Karnataka	Udupi	3	3	0
2020-04-30	Karnataka	Udupi	3	3	0
2020-05-01	Karnataka	Udupi	3	3	0
2020-05-02	Karnataka	Udupi	3	3	0
2020-05-03	Karnataka	Udupi	3	3	0
2020-05-04	Karnataka	Udupi	3	3	0

Table 7.1: Processed COVID-19 Indian districts' time-series dataset

The above mentioned processed dataset is further used to derive the number of daily active cases which can be derived from the following formula

$$\text{active cases} = \text{confirmed cases} - \text{recovered cases} - \text{deceased cases}$$

The above formula is applied over all the rows of every dataframe to obtain the number of active cases on any given day. This record of the number of active cases is utilized to make a forecast on the future number of daily active cases and daily new active cases.

7.2 Data Science Models

The processed dataset of active cases is fed into a statistical model to obtain the desired results. Three such statistical models are implemented and experimented in this study using which forecast is obtained.

The results and the accuracy obtained from tests are compared to determine which model is most suitable for forecasting the pandemic and the term of days that gives the best model results. The three models along with their implementation details are listed below.

Exponential Smoothing

The standard built-in exponential smoothing function is available in Python inside the statsmodels [2] library.

- The dataset is converted to a dataframe with a single column which corresponds to date and number of active cases being the key-value pair.
- The exponential smoothing function is simply called using the preprocessed dataset as shown below

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing  
  
model = ExponentialSmoothing('dataset', trend = 'mul', seasonal = 'mul', seasonal_periods = m_days).fit()  
  
result = model.forecast(n)
```

Autoregressive Integrated Moving Average (ARIMA)

- ARIMA model is a combination of Autoregressive (AR) and Moving Average (MA) models that can predict based on the past values. ARIMA models are widely used in stock market analysis and they can be used over non-stationary data where the dataset is converted to stationary using certain transformation techniques.
- Non-stationary dataset is made stationary by performing differencing, logarithms or square roots of the values.
- The dataset is iteratively checked for stationarity using Augmented Dickey-Fuller (ADF) test. The plot for ADF tests of original dataset and transformed to stationary dataset are shown in figure 7.1.

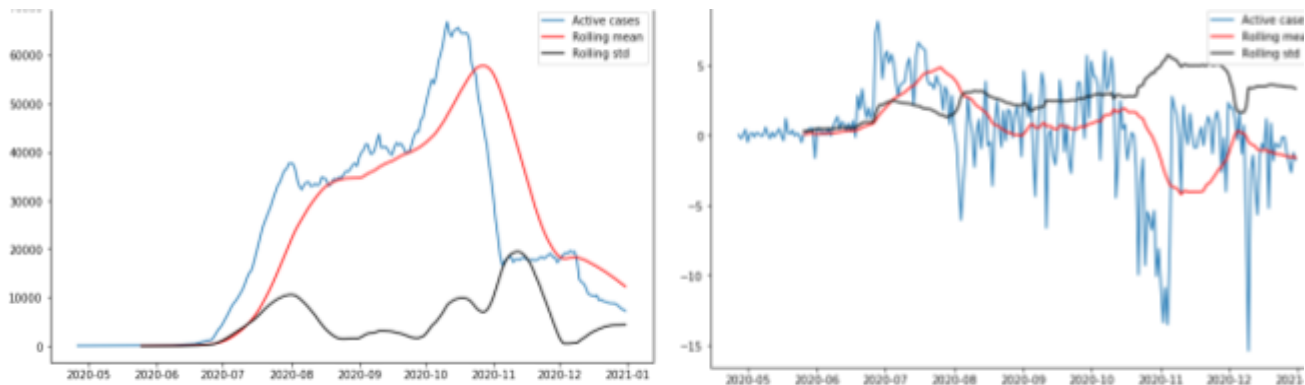


Figure 7.1: Rolling statistics plot to check stationarity of the data

- The AutoCorrelation Function(ACF) and Partial AutoCorrelation Function(PACF) plots shown in figure 7.2 are obtained for the stationary dataset to determine the order of individual AR and MA models.

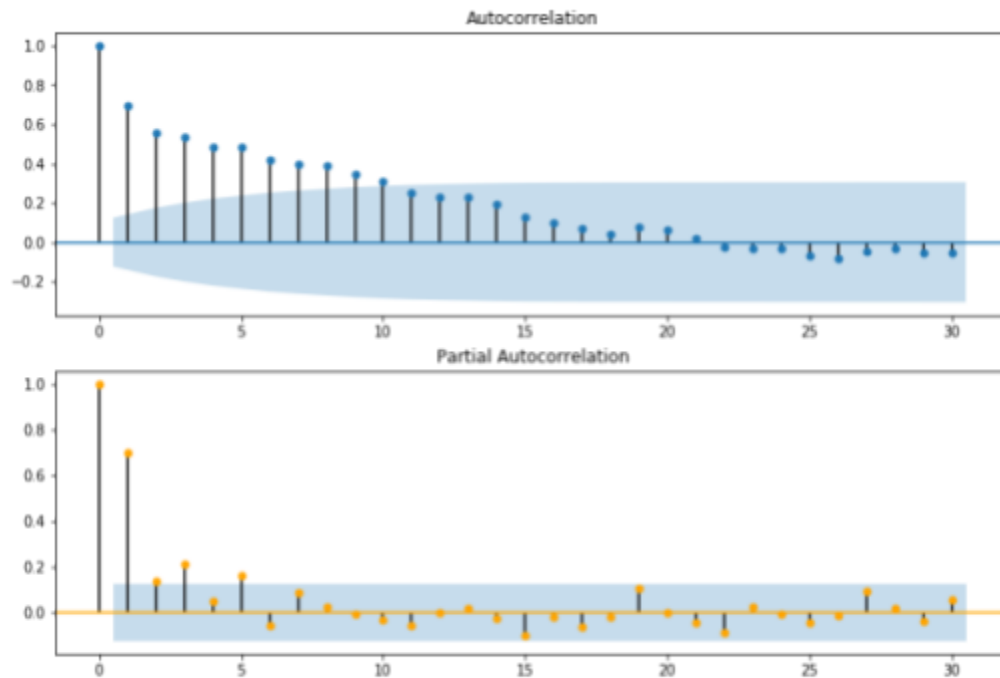


Figure 7.2: ACF and PACF plots to determine p and q value

- Now ARIMA model is implemented by simply calling the built-in ARIMA function which is available as a part of statsmodels library. The order of the fit has to be mentioned as a parameter to the function.
- The forecast of the model can be extracted from the fittedvalues attribute of the model. Transformations are applied over these values to obtain the forecast in the original form.

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing  
  
# stationarity check and transformations  
  
model = ARIMA('dataset', order = (p, d, q))  
results = model.fit()  
forecast = pd.Series(results.fittedvalues, copy=True)  
  
# transform the forecast values back to original form
```

Long Short-term Memory(LSTM)

Recurrent Neural Networks(RNN) are Deep Learning ensembles used for prediction in sequential data. RNN has memory which makes it suitable for time series analysis over a set of past values. They process time series data step-by-step and its internal memory holds the state of the data for each timestamp.

- The time series dataset is divided into a 7-day time frame filter where each time frame would consist of the data of 7 consecutive days and forecast the value for the 8th day based on those days.
- Deep learning models are implemented using Keras[45] library of Python. The RNN model is built up of 2 LSTM layers and a dense layer. The structure of the model is represented in figure 7.3.
- The RNN model is trained for 300 epochs and the forecast is obtained for the desired number of days.

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 7, 50)	10400
lstm_1 (LSTM)	(None, 50)	20200
dense (Dense)	(None, 1)	51
Total params: 30,651		
Trainable params: 30,651		
Non-trainable params: 0		

Figure 7.3: RNN model

```

from tensorflow.keras.models import sequential
from tensorflow.keras.layers import LSTM
# data divided into timeframe of n-days

model = Sequential()
model.add( LSTM( 50, activation = 'relu', return_sequences = True, input_shape = ( n_steps, n_features)))
model.add( LSTM( 50, activation = 'relu'))
model.add(Dense(1))
model.compile(optimizer = 'adam', loss = 'mse')
# fit model
model.fit(Train_data, Train_labels, epochs=300, verbose=1)
results = model.predict(Actual_data)

```

7.3 Test Strategy

The objective of the project is to utilize a year long data of the pandemic spread to detect the number of cases and as of the day this report is framed the size of the dataset available is over 11 months of data. The data is divided into two parts as data from 2020 and data from 2021. The models are tested on two different datasets; firstly the dataset of 2020 and the whole dataset except for the last 15 days from when the forecast is to be made.

First test interval: 2021-01-01 to 2021-01-15

Second test interval: 2021-04-13 to 2021-04-20

The dataset of year 2020 is further divided quarterly and different lengths of dataset is derived by combining the quarterly data. This data is used to determine what volume of data gives better results. The results obtained during both the timeframes are tallied with the actual number of COVID cases and the error is determined. This error is standardized and compared with the other models to see the performance of each of them. Error is calculated using the following 2 metrics

Root mean square error

$$RMSE = \frac{\sqrt{\sum (x_i - \hat{x}_i)^2}}{N}$$

Root mean square percentage error

$$RMPSE = \frac{\sqrt{\sum_i \left(\frac{x_i - \hat{x}_i}{x_i^2} \right)^2}}{N} * 100$$

The optimum number of days for which the best forecast can be obtained is determined by comparing the forecast result accuracy and considering the confidence interval of the forecast of the above mentioned time frame. The forecast is thus obtained for the next n days which is determined to be optimum.

CHAPTER 8

RESULTS AND DISCUSSION

Forecast over the number of COVID-19 cases by any model has to be as accurate as possible and hence the forecast has to be obtained with good accuracy and for a respectable time period such that the results can give insights over the situation. The basic belief of forecast is that the accuracy of the result decreases as the volume of forecast increases as uncertainty of prediction increases with the time frame of forecast. This shall be examined to determine how many days of forecast would give accurate results.

The results illustrated in this section is by performing the forecast over the data detailed in dataset design(section 6.2) and the models shown in low level design(section 6.3). The results are obtained and displayed for the district of 'Bengaluru Urban' and discussion is made to draw conclusions from the observations.

The forecast power of a model depends highly on the dataset. Hence an analysis is performed to determine what volume of data gives best results.

A quarter-wise analysis of the dataset is performed by dividing the 2020 dataset quarterly into 3

	Date	Quarter 4 data	Quarter 4 and 3 data	All data
Time frame				
1	2021-01-01	97.75%	95.78%	97.59%
2	2021-01-02	98.15%	96.99%	97.68%
3	2021-01-03	98.03%	96.18%	97.80%
4	2021-01-04	98.16%	95.96%	97.91%
5	2021-01-05	97.75%	96.38%	97.77%
6	2021-01-06	95.70%	95.92%	96.37%
7	2021-01-07	95.46%	95.56%	96.21%
8	2021-01-08	95.71%	95.84%	96.45%
9	2021-01-09	95.94%	95.70%	96.61%
10	2021-01-10	95.76%	94.54%	96.19%
11	2021-01-11	95.12%	93.07%	95.17%
12	2021-01-12	94.88%	92.26%	94.69%
13	2021-01-13	94.34%	91.35%	93.79%
14	2021-01-14	93.67%	90.79%	92.93%

Table 8.1: Quarterwise analysis of the dataset

quarters. The models are then tested with these smaller datasets to determine as to what volume of data yields best results as illustrated in table 8.1.

It can be observed that the fourth quarter alone gives best results and this anomaly can be justified due to the specific trend of COVID-19 pandemic spread in Bengaluru Urban district using the Exponential smoothing model and this case is not true for all the other districts. Since the accuracy of the forecast using the whole dataset is as accurate as the last quarter alone and the fact that the whole dataset provides more knowledge about the condition, it can certainly be concluded that larger amounts of data gives better accuracy.

The 2020 dataset is used to forecast the number of COVID-19 cases for the first 15 days of the year 2021. This forecast result is compared with the actual number of cases in Bengaluru Urban during that time and the performance metric is derived.

Date	Forecast	Daily forecast	Confidence interval	Actual	Daily
2021-01-01	6992.0	-242	98.68%	7086	-148
2021-01-02	6800.0	-192	97.84%	6965	-121
2021-01-03	6626.0	-174	95.68%	7123	158
2021-01-04	6468.0	-158	95.90%	6692	-431
2021-01-05	6325.0	-143	96.11%	6148	-544
2021-01-06	6196.0	-129	94.87%	5680	-468
2021-01-07	6079.0	-117	94.34%	5623	-57
2021-01-08	5973.0	-106	94.66%	5860	237
2021-01-09	5878.0	-95	94.97%	5877	17
2021-01-10	5793.0	-85	95.02%	6069	192
2021-01-11	5716.0	-77	95.07%	5975	-94
2021-01-12	5647.0	-69	95.28%	5640	-335
2021-01-13	5585.0	-62	95.44%	5679	39
2021-01-14	5530.0	-55	95.61%	5562	-117

The forecast is performed with all the 3 models while the results obtained using the ARIMA model and the Exponential smoothing models are illustrated in figures 8.1 and 8.2 respectively. The forecast is obtained for 14 days from the start of the year 2021.

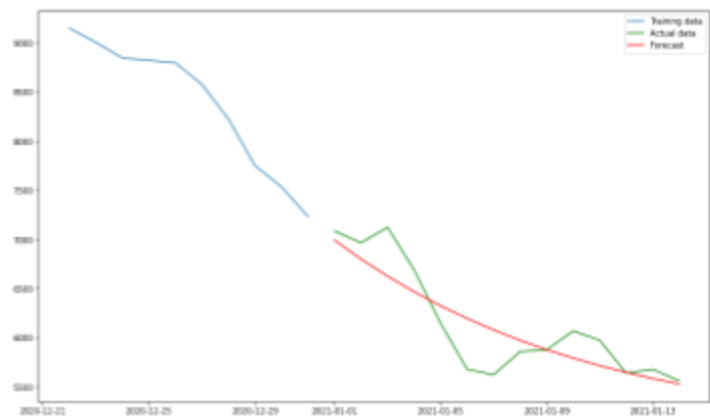


Figure 8.1: Forecast over the first test interval by ARIMA model

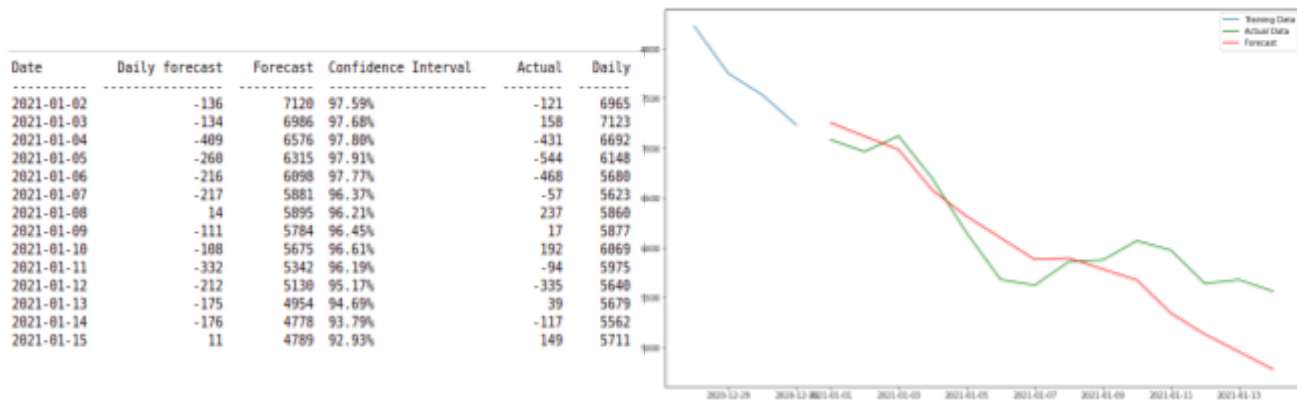


Figure 8.2: Forecast over the first test interval by ES model

It can be seen from the graph and forecast values that the forecast obtained by the ARIMA model and ES models over the first test interval has impressive accuracy. While the ARIMA model has a better prediction power for a longer term than ES, ES gives much more accurate results for a shorter forecasting period. However LSTM performed the worst with accuracy of 75% for a forecast of 15 days.

The data from 2020 and 2021 are combined to obtain the complete COVID-19 data over the year to forecast using yearlong data. The models are further tested by using the whole dataset by forecasting pandemic spread for 15 days before the current date. This time the test is performed to determine what window of forecast gives best accuracy and thus the future forecast can be confidently obtained for that time period.

Date	Daily	Cumulative	Confidence Interval	Daily	Cumulative
2021-04-05	1952	31579	97.59%	2684	30783
2021-04-06	2160	33739	97.68%	1823	32606
2021-04-07	2536	36275	97.80%	3184	35790
2021-04-08	2996	39272	97.91%	3157	38947
2021-04-09	4040	43313	97.77%	3579	42526
2021-04-10	5073	48386	96.37%	2338	44864
2021-04-11	5406	53793	96.21%	6373	51237
2021-04-12	3544	57338	96.45%	5309	56546

Date	Daily	Cumulative	Confidence Interval	Daily	Cumulative
2021-04-05	3081	33936	97.0886	1823	32606
2021-04-06	2018	35947	93.6687	3184	35790
2021-04-07	2342	38290	94.8939	3157	38947
2021-04-08	3272	41562	82.3734	3579	42526
2021-04-09	3229	44792	89.4064	2338	44864
2021-04-10	2652	47445	91.6851	6373	51237
2021-04-11	3157	50603	93.6864	5309	56546
2021-04-12	3825	54428	77.6367	0	56546

Figure 8.3: Forecast over second test interval by ES and LSTM models

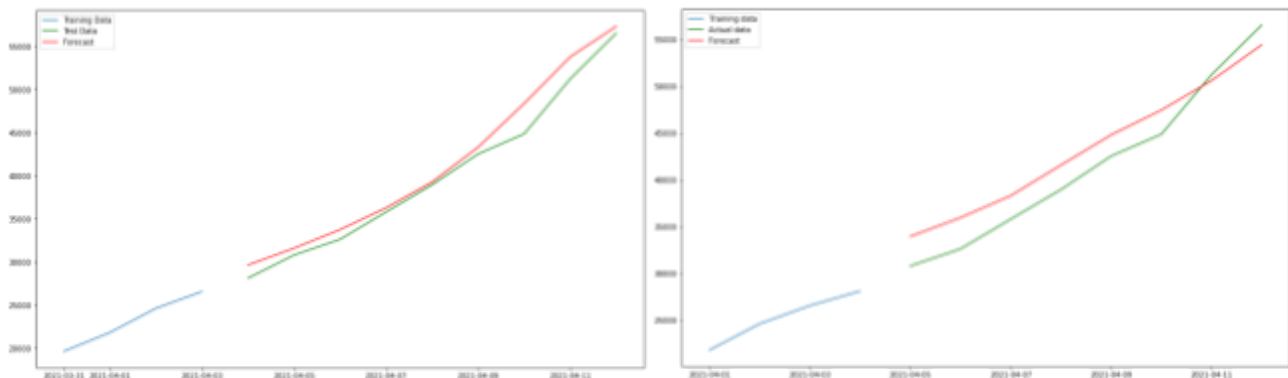


Figure 8.4: Forecast plots over second test interval by ES and LSTM models

The tests are performed by forecasting COVID iteratively for the last 15 days of the dataset and the forecast results are obtained for all the 3 models. The results obtained from the Exponential smoothing model and LSTM model are displayed in figure 8.3 and 8.4 respectively. From the experimental results it was found that a forecast for 7 to 9 days would yield very accurate results. Based on these assumptions, the forecast is obtained for the next 8 days over the Bengaluru Urban district using all 3 models and their results are depicted in figure 8.5, 8.6 and 8.7.

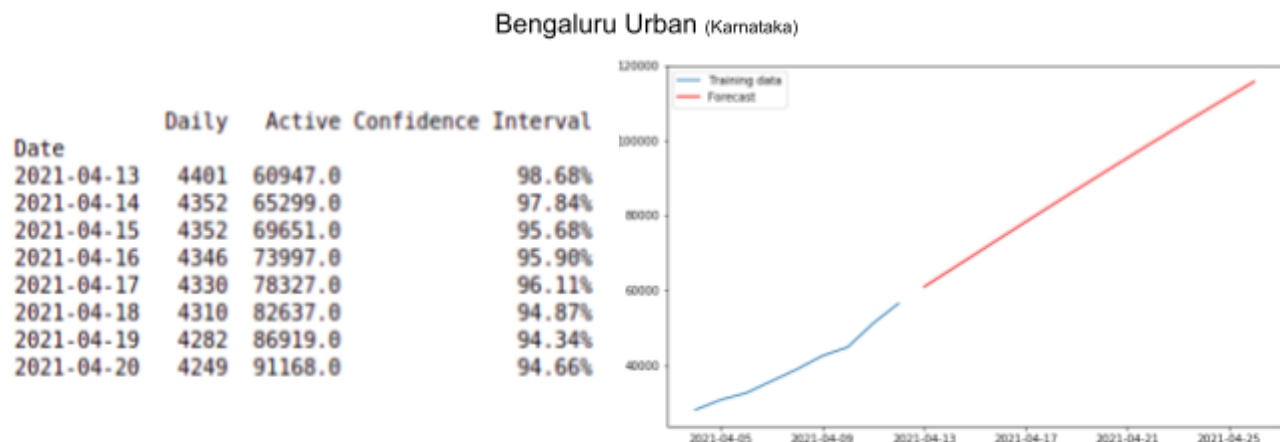


Figure 8.5: Future forecast of pandemic over Bengaluru Urban district by ARIMA model

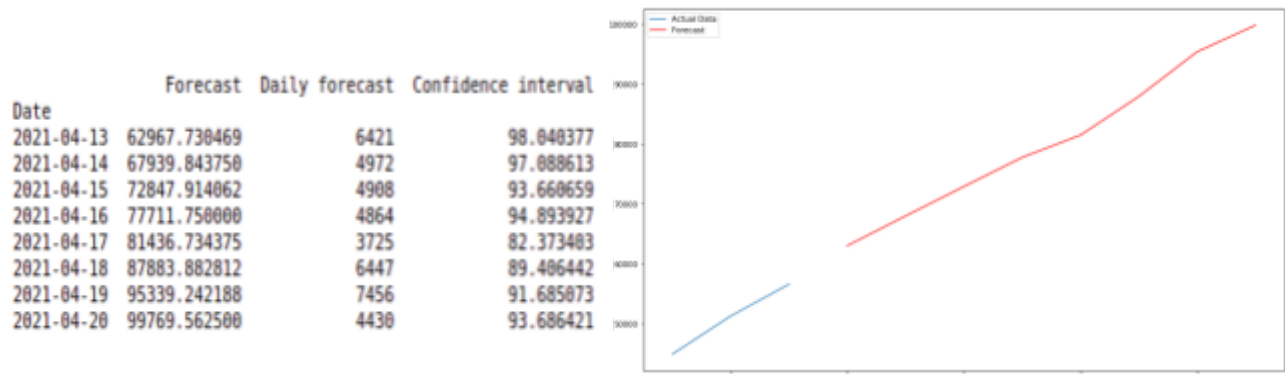


Figure 8.6: Future forecast of pandemic over Bengaluru Urban district by LSTM model

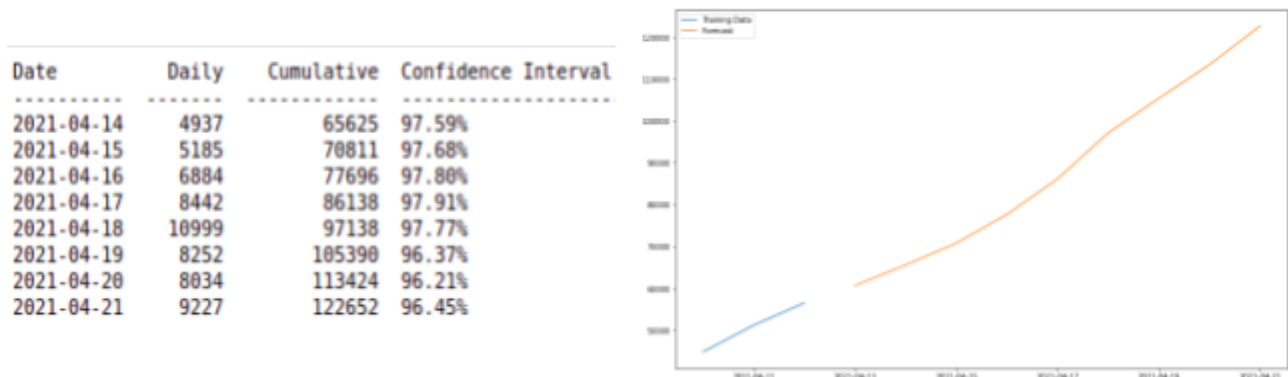


Figure 8.7: Future forecast of pandemic over Bengaluru Urban district by ES model

It can be observed that the ARIMA model and Exponential smoothing model exhibit higher accuracy in comparison to the LSTM model. The results state that the number of COVID cases would increase over a period of the next few days. While the Exponential smoothing model shows a very large spike in the number of cases the ARIMA model represents a steady growth in the number of COVID cases. The forecast accuracy of the 3 models in comparison is

$$\text{ARIMA} \geq \text{Exponential Smoothing} \gg \text{LSTM}$$

The trend of COVID-19 in other parts of the country can as well be obtained from these models. The Exponential Smoothing model seems to provide a good short-term fit and hence the ES model is concluded to be the best forecast model and forecast results of certain other districts as well are illustrated using the ES model.

The spread of COVID-19 virus over the district of Chittoor, Andhra Pradesh forecasted by ES model is visualized in figure 8.8 and 8.9.

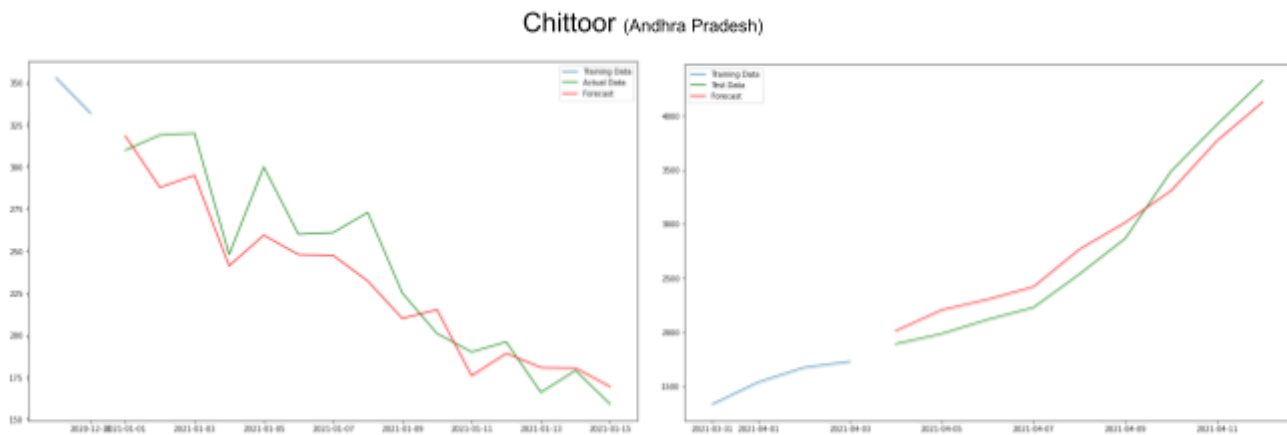


Figure 8.8: Forecast plots of Chittoor district over first and second test interval by ES models

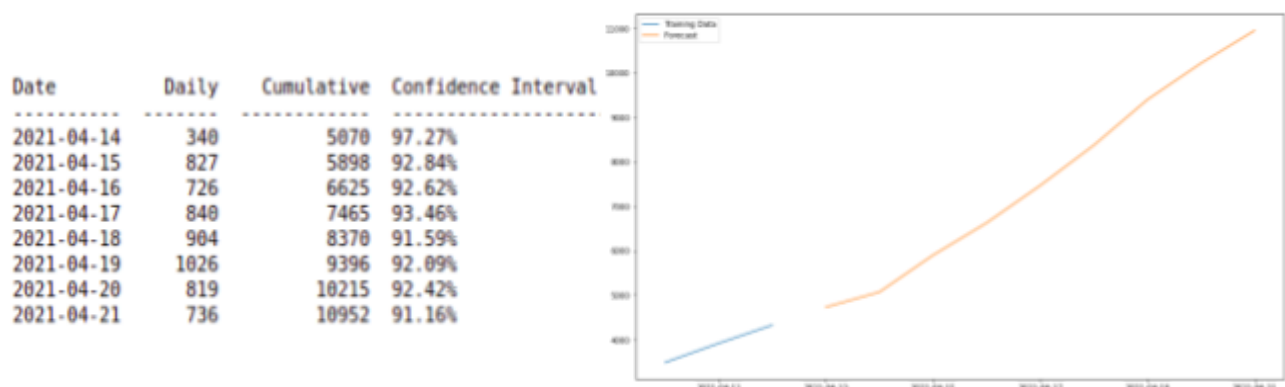


Figure 8.9: Future forecast of pandemic over Chittoor district by ES model

Similarly the pandemic growth is studied and analysed for Kolkata district of West Bengal state. The trend of the pandemic spread in the Kolkata district is illustrated in figure 8.10 and 8.11.

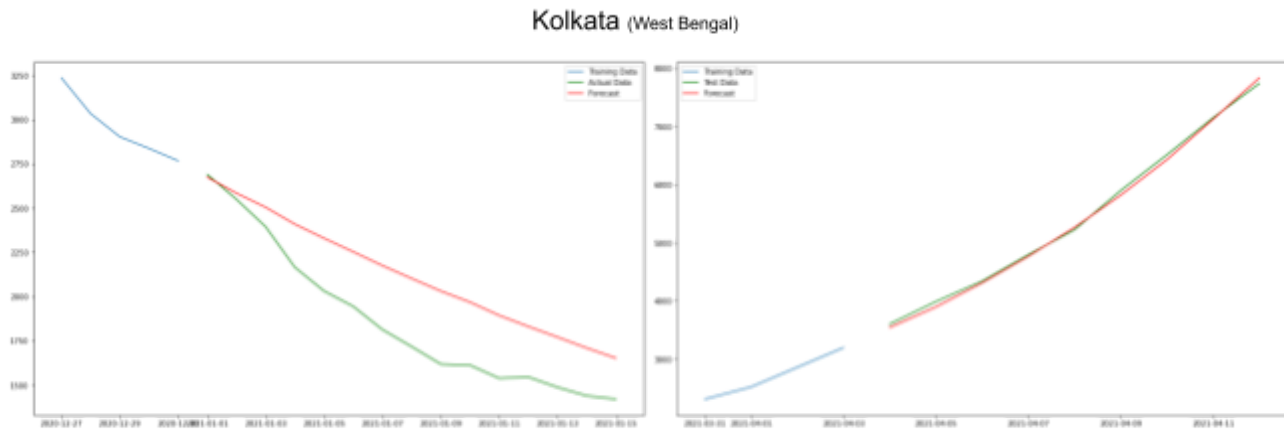


Figure 8.10: Forecast plots of Kolkata district over first and second test interval using ES models

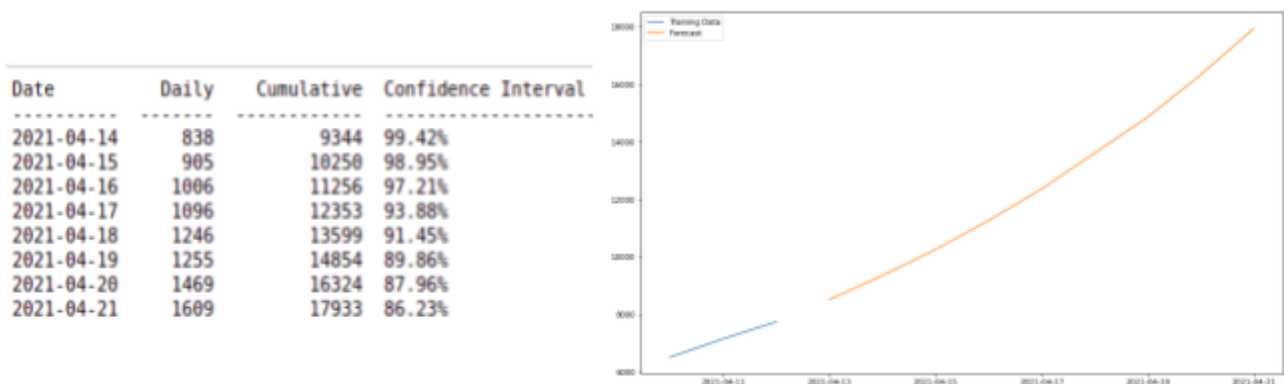


Figure 8.11: Future forecast of pandemic over Kolkata district by ES model

The results show a steep increase of COVID-19 in the above 3 districts among which Bengaluru Urban is forecasted to have the most cases. Similarly all the other districts of India are going to witness a rise in COVID-19 cases. While the effects of COVID-19 virus were seen in the first wave, the second wave can only be predicted to be far more devastating.

CHAPTER 9

CONCLUSIONS AND FUTURE WORK

9.1 Conclusions

Usage of statistical models for forecasting pandemic is witnessed before and this study further supports the fact that data science can be efficiently utilized to determine the future state of a pandemic. The study successfully illustrates the implementation of statistical models to forecast the spread of COVID-19 pandemic.

The results of forecasting using the ARIMA, Exponential Smoothing and LSTM models are illustrated in the results and discussion section(section 8). We observe that the forecasting depends more over the data used than the models implemented and the dataset has to be accurate and consistent. Good forecast results can always be obtained with adequate data and limited data often lead to inaccuracies in the results.

A model can confidently forecast the future but for a limited amount of time due to the uncertainties especially over non-seasonal data. Hence the model in this study can forecast the number of cases for 7 to 9 days. All the implemented models give varied results with different accuracies and comparing their performances, conclusions can be made that the ARIMA model and Exponential smoothing model give better forecast results. The accuracies of both ARIMA

and ES models are similar while that of the LSTM model is poor in comparison. Hence the forecast by Exponential Smoothing is concluded as the best forecast.

COVID-19 pandemic spread does not exhibit seasonal changes and varies very unpredictably without a pattern. This feature makes the virus spread very difficult to be forecasted and hence it is not suitable for long term forecast. The spread does not seem to be affected by the climate and increases with increase in the density of population.

The model represents a rise in the number of cases for the next few days and confidently states an exponential increase in the number of cases for the next 10 days. The ES model further states that the condition of the pandemic spread would worsen in the country while the implemented forecast models foresee the number of COVID cases hitting the mark of 1,00,000 in Bengaluru Urban district by the next 6 days i.e 19th April 2021.

Relying on the forecast results obtained from the model, a conclusion can be made that the country would witness a huge number of COVID-19 cases which can be accounted as the second wave of the pandemic. The effects of the pandemic spread can be countered or atleast lowered by enforcing lockdown and increasing the hospital and medical capacities in the country.

9.2 Future Work

Forecasting is habitually used in market-forecasting and other applications. Statistical models can be used for forecasting any pandemic in the future. The implemented model can address the influence of other parameters on the spread of the pandemic to improve the forecast

accuracy and the relation between those parameters can be studied. The implemented COVID-19 forecast models however can be further utilized to study the effect of the pandemic over other parameters such as stock market, economic index, happiness index, etc.

Further these models can be utilized to make forecasts of other countries as well and insights obtained from the pandemic spread in India can be used. The observations can be used in the future if such a pandemic were to occur again.

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APPENDIX A DEFINITIONS, ACRONYMS AND ABBREVIATIONS

Abbreviations / Acronyms

SARS-CoV - 2	:	Severe acute respiratory syndrome coronavirus 2
WHO	:	World Health Organization
ML	:	Machine Learning
LR	:	Linear Regression
LSTM	:	Long-short Term Memory
MA	:	Moving Average
LASSO	:	Least Absolute Shrinkage Selection Operator
SVM	:	Support Vector Machines
ES	:	Exponential Smoothing
MSE	:	Mean Square error
RMSE	:	Root Mean Square error
RMPSE	:	Root Mean Square Percentage error
MAE	:	Mean Absolute error
MAPE	:	Mean Absolute Percentage error
IIT	:	Indian Institute of Technology
GiB	:	Gibibyte
SSD	:	Solid State Drive
ARIMA	:	Auto Regressive Integrated Moving Average
ADF	:	Augmented Dickey-Fuller
AIC	:	Akaike Information Criterion

ACF	:	AutoCorrelation Function
PACF	:	Partial AutoCorrelation Function
RNN	:	Recurrent Neural Network
DNA	:	Data Not Available
SAARC	:	South Asian Association for Regional Cooperation
NLP	:	Natural Language Processing
SEIR	:	Susceptible - Exposed - Infected - Recovered
GB	:	Gigabyte
DDR	:	Double data rate
RAM	:	Random Access Memory